Data-Backed Journal-Style Replication of *Something Old, Something New*

Codex journal-replication workflow

2026-04-04

Abstract

This manuscript expands the local replication folder for Katila and Ahuja (2002) into a long journal-style paper using only processed datasets already present in the workspace. The current local batch contains 2,174 observations across 162 entities, and the batch headline estimate is 0.0052 for search_scope with p = 0.6153. The local quantity model points to a nonlinear depth effect and a weaker role for search scope than the source paper emphasizes. The draft imitates the structure of the source article while documenting the exact places where local replication decisions depart from direct source facts.

Keywords: search depth; search scope; new product introduction; firm-year innovation; local analogue

1 Paper-First Reading Protocol

The workflow for this manuscript is intentionally ordered. The source paper is read first, the required empirical objects are identified second, the local files are verified third, and only then is the replication paper written from the checked evidence chain.

Step	Action	Evidence
1	Read the original paper before touching the local estimates.	C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/KatilaAhujaamj02.txt
2	Extract the source paper's theoretical logic, sample definition, variable definitions, and estimand.	Something Old, Something New: A Longitudinal Study of Search Behavior and New Product Introduction
3	Match each required empirical object to a local executable file and verify that the file exists.	C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/katila-ahuja-2002-search-behavior/results/katila_ahuja_2002_analysis_dataset.cs — C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/katila-ahuja-2002-search-behavior/results/katila_ahuja_2002_model_coefficients — C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/katila-ahuja-2002-search-behavior/results/katila_ahuja_2002_sample_summary.c

Step	Action	Evidence
4	Write the replication paper only from the checked local files and calculated outputs.	The manuscript, markdown notes, and PDF are generated from the verified file set.

Ordered protocol used for this replication paper.

2 Required Data and Local File Manifest

This section answers the basic replication question of where the data are. Each empirical object is mapped to a concrete local file before the estimates are interpreted. Workspace root: C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions

Stage	NeededFor	FilePath	Exists	Status
Source article	Recover theory, hypotheses, sample rules, definitions, tables, notes, and appendices	patent literature/KatilaAhujaamj02.txt	Yes	Direct source
Analysis panel	Estimate the local model and regenerate descriptive statistics	patent literature/daily_replications/2026-04-04/katila-ahuja-2002-search-behavior/results/katila_ahuja_2002_analysis_dataset.csv	Yes	Executable local dataset

Stage	NeededFor	FilePath	Exists	Status
Coefficient export	Recover the reported local coefficients and p-values	patent literature/daily_replications/2026-04-04/katila-ahuja-2002-search-behavior/results/katila_ahuja_2002_model_coefficients.csv	Yes	Executable local result file
Sample summary	Audit sample construction and raw grouped moments	patent literature/daily_replications/2026-04-04/katila-ahuja-2002-search-behavior/results/katila_ahuja_2002_sample_summary.csv	Yes	Executable local result file

Exact local files used by the manuscript.

3 Introduction

The source paper asks whether search depth and search scope are distinct dimensions and how they shape subsequent innovation. The point of the present draft is not to pretend that every original source file has been recovered. It is to turn the executable local evidence into a paper that preserves theory, data architecture, variable mapping, and empirical claims in one place. The current local batch contains 2,174 observations across 162 entities, and the batch headline estimate is 0.0052 for search_scope with $p = 0.6153$. The local quantity model points to a nonlinear depth effect and a weaker role for search scope than the source paper emphasizes.

4 Research Positioning and Contribution

The replication matters because the original paper's distinction between search depth and search scope is widely cited, yet the empirical mechanism depends heavily on how those constructs are operationalized. The paper's contribution is to recover the two-dimensional search logic in an executable patent panel and show where the patent-based analogue departs from the original product-outcome result. The writing stance follows a conservative paper-first workflow: the source paper defines the theoretical target, the local files define the executable boundary, and the replication paper names the distance between the two rather than smoothing it over.

5 Original Paper, Theory, and Published Estimand

The focal source article is [1]. The source paper is Something Old, Something New: A Longitudinal Study of Search Behavior and New Product Introduction by Riitta Katila and Gautam Ahuja in Academy of Management Journal . The original paper distinguishes search depth from search scope. The study is longitudinal and firm-year based. Innovation quantity and quality are both part of the argument. A useful replication discipline is to keep source facts and replication decisions separate. Source facts describe what the paper says. Replication decisions describe what the local executable workflow must do because it is limited to processed datasets already present in the workspace.

- The original paper distinguishes search depth from search scope.
- The study is longitudinal and firm-year based.
- Innovation quantity and quality are both part of the argument.

DirectSourceFact	LocalReplicationDecision
Depth and scope are distinct dimensions.	The local panel maps depth and scope to executable recombination constructs.
The original design is firm-year based.	The firm-year structure is preserved exactly.
Innovation outcomes include quantity and quality.	Patent counts and patent-impact totals substitute for product introductions.

Direct source facts are kept separate from local replication decisions.

6 Decision Register

The table below makes the major design choices explicit before any coefficient is interpreted.

DecisionPoint	LocalSpecification
Research question	The source paper asks whether search depth and search scope are distinct dimensions and how they shape subsequent innovation.
Unit of analysis	Firm-year
Sample frame	Firm-year panel aggregated from the local patent-level recombination dataset for firms with usable year-to-year transitions.

DecisionPoint	LocalSpecification
Time window	1986 to 2015 in the local executable panel.
Outcome definition	Next-year patent quantity and next-year total patent impact.
Main explanatory design	Firm and year fixed-effects models relating next-period innovation to current search depth, scope, and their interaction.
Controls and fixed effects	Current innovation scale and field-experience controls with firm and year effects.
Exclusion or substitution logic	The local executable design substitutes patent-based future outcomes for product-introduction outcomes and keeps only firm-years with usable lag-lead construction.

Decision register for the local replication design.

7 Local Data Inventory and Sample Reconstruction

The local analogue aggregates the patent-level recombination panel to firm-years and uses patent-based outcomes in place of the source paper's product-introduction outcomes. Every number in

this manuscript is backed by local files in patent literature/daily_replications/2026-04-04/katila-ahuja-2002-search-behavior or by the explicitly referenced processed regression panel.

- Firm-year panel aggregated from the local patent-level recombination dataset.
- Search depth measured from prior organizational recombination experience.
- Search scope measured from breadth of recombination in the current year.

File	Role
katila_ahuja_2002_analysis_dataset.csv	Firm-year search-behavior panel
katila_ahuja_2002_model_coefficients.csv	Quantity and impact coefficient table
katila_ahuja_2002_sample_summary.csv	Sample statistics

Local executable files used by this paper.

Statistic	Value
rows	2174.0000
unique_firms	162.0000
year_min	1986.0000
year_max	2015.0000
mean_search_depth	0.6637
mean_search_scope	4.2347
mean_next_patents	10.6026

Sample reconstruction table from the local results folder.

8 Variable Mapping, Derivations, and Replication Decisions

The variable map is the bridge between the source article and the executable local workflow.

OriginalConstruct	LocalMeasure	ExactVariable	Derivation
Search depth	Mean prior organizational experience	search_depth	Average the current-year patent-level organizational experience within GVKEY-year.
Search scope	Mean breadth	search_scope	Average current-year breadth within GVKEY-year.
Current innovation scale	Current-year patent count	current_patents	Count current-year patents.
Next-period quantity	Next-year patent count	next_patents	Shift the patent count one year forward within GVKEY.
Next-period impact	Next-year total impact	next_total_impact	Shift the impact total one year forward within GVKEY.

Variable mapping from source constructs to local measures. Firm and year fixed-effects models for next-year patent counts and next-year total impact using local depth, scope, and their interaction.

$$E[\text{NextPatents}_{it} \mid X_{it}] = \exp\left(\beta_1 \text{Depth}_{it} + \beta_2 \text{Depth}_{it}^2 + \beta_3 \text{Scope}_{it} + \beta_4 \text{Depth}_{it} \times \text{Scope}_{it} + \alpha_i + \gamma_t\right), \tag{1}$$

$$\log(1 + \text{NextImpact}_{it}) = \theta_1 \text{Depth}_{it} + \theta_2 \text{Depth}_{it}^2 + \theta_3 \text{Scope}_{it} + \theta_4 \text{Depth}_{it} \times \text{Scope}_{it} + \alpha_i + \gamma_t + \varepsilon_{it}. \tag{2}$$

9 Empirical Design and Identification Limits

The local design remains strongest when it preserves the original unit of analysis and interaction logic, even if some measurement layers are reconstructed with defensible analogues rather than identical source files.

- The local analogue uses patent outcomes rather than product-introduction outcomes.
- Search variables are mapped from recombination measures already in the workspace.
- The exercise is strongest on conceptual mechanism and panel structure, not exact outcome identity.

10 Main Results

Search depth is estimated at -0.2243 with p = 0.0250 in model katila_ahuja_next_patents. Search depth squared is estimated at 0.1059 with p = ¡0.001 in model katila_ahuja_next_patents. Search scope is estimated at 0.0052 with p = 0.6153 in model katila_ahuja_next_patents. Log(1 + current patents) is estimated at 0.6654 with p = ¡0.001 in model katila_ahuja_next_patents. Mean field experience is estimated at -0.0429 with p = 0.1018 in model katila_ahuja_next_patents. Depth x scope is estimated at -0.0115 with p = 0.2651 in model katila_ahuja_next_patents. Search depth is estimated at -0.1070 with p = 0.3854 in model katila_ahuja_next_impact. Search depth squared is estimated at 0.0388 with p = 0.2375 in model katila_ahuja_next_impact. Search scope is estimated

at 0.0184 with p = 0.1671 in model katila_ahuja_next_impact. Log(1 + current patents) is estimated at 0.6357 with p = ¡0.001 in model katila_ahuja_next_impact. Mean field experience is estimated at 0.0271 with p = 0.4444 in model katila_ahuja_next_impact. Depth x scope is estimated at -0.0216 with p = 0.0982 in model katila_ahuja_next_impact.

model	Term	estimate	std_error	p_value
katila_ahuja_next_patents	Search depth	-0.2243	0.1001	0.0250
katila_ahuja_next_patents	Search depth squared	0.1059	0.0299	0.0004
katila_ahuja_next_patents	Search scope	0.0052	0.0104	0.6153
katila_ahuja_next_patents	Log(1 + current patents)	0.6654	0.0652	0.0000
katila_ahuja_next_patents	Mean field experience	-0.0429	0.0262	0.1018
katila_ahuja_next_patents	Depth x scope	-0.0115	0.0103	0.2651
katila_ahuja_next_impact	Search depth	-0.1070	0.1214	0.3854
katila_ahuja_next_impact	Search depth squared	0.0388	0.0322	0.2375
katila_ahuja_next_impact	Search scope	0.0184	0.0130	0.1671
katila_ahuja_next_impact	Log(1 + current patents)	0.6357	0.0617	0.0000
katila_ahuja_next_impact	Mean field experience	0.0271	0.0350	0.4444
katila_ahuja_next_impact	Depth x scope	-0.0216	0.0126	0.0982

Key coefficient table used in the main text.

Term	Model	Estimate	PValue	Signal	Reading
Search depth	katila_ahuja_next_patents	-0.2243	0.0250	Moderate	Negative estimate with p = 0.0250.

Term	Model	Estimate	PValue	Signal	Reading
Search depth squared	katila_ahuja_next_patents	0.1059	0.0004	Strong	Positive estimate with p = ¡0.001.
Search scope	katila_ahuja_next_patents	0.0052	0.6153	Weak	Positive estimate with p = 0.6153.
Log(1 + current patents)	katila_ahuja_next_patents	0.6654	0.0000	Strong	Positive estimate with p = ¡0.001.
Mean field experience	katila_ahuja_next_patents	-0.0429	0.1018	Weak	Negative estimate with p = 0.1018.
Depth x scope	katila_ahuja_next_patents	-0.0115	0.2651	Weak	Negative estimate with p = 0.2651.
Search depth	katila_ahuja_next_impact	-0.1070	0.3854	Weak	Negative estimate with p = 0.3854.
Search depth squared	katila_ahuja_next_impact	0.0388	0.2375	Weak	Positive estimate with p = 0.2375.

Term	Model	Estimate	PValue	Signal	Reading
Search scope	katila_ahuja_next_impact	0.0184	0.1671	Weak	Positive estimate with p = 0.1671.
Log(1 + current patents)	katila_ahuja_next_impact	0.6357	0.0000	Strong	Positive estimate with p = ¡0.001.
Mean field experience	katila_ahuja_next_impact	0.0271	0.4444	Weak	Positive estimate with p = 0.4444.
Depth x scope	katila_ahuja_next_impact	-0.0216	0.0982	Suggestive	Negative estimate with p = 0.0982.

Interpretive summary of the main coefficients.

Variable	N	Mean	SD	P25	Median	P75	Min	M
Search depth	2174.0000	0.6637	0.6622	0.1386	0.5493	0.9635	0.0000	4.76
Search scope	2174.0000	4.2347	2.8896	2.8182	3.4264	4.6667	2.0000	52.00
Current patents	2174.0000	10.5437	20.1122	2.0000	5.0000	11.0000	1.0000	399.00
Current total impact	2174.0000	56.5745	142.3809	5.0000	17.0000	47.0000	0.0000	3058.00
Mean field experience	2174.0000	1.4705	1.1482	0.6931	1.3041	1.9634	0.0000	7.55
Next-year patents	2174.0000	10.6026	21.2666	2.0000	4.0000	11.0000	1.0000	399.00

Variable	N	Mean	SD	P25	Median	P75	Min	M
Next-year total impact	2174.0000	56.6297	144.9651	5.0000	16.0000	46.0000	0.0000	3058.00

Descriptive statistics for the main variables.

Quantile	N	MeanRegressor	MeanOutcome
1.0000	870.0000	0.1128	5.3149
2.0000	506.0000	0.5658	11.7273
3.0000	363.0000	0.8803	17.4105
4.0000	435.0000	1.6987	14.1885

Quantile summary linking the main regressor to the local outcome.

11 Claim-Evidence Audit

This audit prevents the paper from blurring together original-paper statements, local estimates, and replication judgments.

Claim	EvidenceType	EvidenceFile	AuditNote
The source paper studies: The source paper asks whether search depth and search scope are distinct dimensions and how they shape subsequent innovation.	Direct source fact	patent literature/KatilaAhujaamj02.txt	Recovered from the source article before local writing begins.

Claim	EvidenceType	EvidenceFile	AuditNote
The local executable panel contains 2,174 observations across 162 entities.	Local calculated fact	patent literature/daily_replications/2026-04-04/katila-ahuja-2002-search-behavior/results/katila_ahuja_2002_analysis_dataset.cs — patent literature/daily_replications/2026-04-04/katila-ahuja-2002-search-behavior/results/katila_ahuja_2002_sample_summary.c	Backed by the verified analysis panel and sample audit.
The local headline estimate is 0.0052 for search_scope with p = 0.6153.	Local calculated fact	patent literature/daily_replications/2026-04-04/katila-ahuja-2002-search-behavior/results/katila_ahuja_2002_model_coefficients	Backed by the exported coefficient file used in the paper.
The manuscript uses local substitutions or analogues wherever the original data architecture is not fully recoverable.	Replication decision	Source-vs-decision table and limitations section	Used only when the original pipeline cannot be matched byte-for-byte from current local files.

Claim-evidence audit for the current manuscript.

12 Comparison to the Published Paper and Remaining Gaps

The local analogue preserves the paper's dual-dimension logic but does not produce the same relative strength for depth and scope. The quantity model highlights a nonlinear depth pattern,

while search scope is comparatively muted in the executable patent-based panel. The honest replication standard is to describe the gap without smoothing it over. A useful local paper tells the next researcher exactly which parts of the original design are recoverable now and which parts remain approximated.

13 Discussion and Conclusion

The manuscript turns a replication folder into a paper that can be uploaded, cited internally, and rerun later without guessing how the evidence was assembled. For Katila and Ahuja (2002) the bottom line is: The local quantity model points to a nonlinear depth effect and a weaker role for search scope than the source paper emphasizes.

A Appendix A: Source Inventory

Stage	NeededFor	FilePath	Exists	Status
Source article	Recover theory, hypotheses, sample rules, definitions, tables, notes, and appendices	patent literature/KatilaAhujaamj02.txt	Yes	Direct source
Analysis panel	Estimate the local model and regenerate descriptive statistics	patent literature/daily_replications/2026-04-04/katila-ahuja-2002-search-behavior/results/katila_ahuja_2002_analysis_dataset.csv	Yes	Executable local dataset
Coefficient export	Recover the reported local coefficients and p-values	patent literature/daily_replications/2026-04-04/katila-ahuja-2002-search-behavior/results/katila_ahuja_2002_model_coefficients.csv	Yes	Executable local result file
Sample summary	Audit sample construction and raw grouped moments	patent literature/daily_replications/2026-04-04/katila-ahuja-2002-search-behavior/results/katila_ahuja_2002_sample_summary.csv	Yes	Executable local result file

Source and local file manifest for the manuscript.

B Appendix B: Reading Protocol

Step	Action	Evidence
1	Read the original paper before touching the local estimates.	C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/KatilaAhujaamj02.txt
2	Extract the source paper's theoretical logic, sample definition, variable definitions, and estimand.	Something Old, Something New: A Longitudinal Study of Search Behavior and New Product Introduction
3	Match each required empirical object to a local executable file and verify that the file exists.	C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/katila-ahuja-2002-search-behavior/results/katila_ahuja_2002_analysis_dataset.cs — C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/katila-ahuja-2002-search-behavior/results/katila_ahuja_2002_model_coefficients — C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/katila-ahuja-2002-search-behavior/results/katila_ahuja_2002_sample_summary.c

Step	Action	Evidence
4	Write the replication paper only from the checked local files and calculated outputs.	The manuscript, markdown notes, and PDF are generated from the verified file set.

Paper-first protocol followed before writing the manuscript.

C Appendix C: Decision Register

DecisionPoint	LocalSpecification
Research question	The source paper asks whether search depth and search scope are distinct dimensions and how they shape subsequent innovation.
Unit of analysis	Firm-year
Sample frame	Firm-year panel aggregated from the local patent-level recombination dataset for firms with usable year-to-year transitions.
Time window	1986 to 2015 in the local executable panel.
Outcome definition	Next-year patent quantity and next-year total patent impact.
Main explanatory design	Firm and year fixed-effects models relating next-period innovation to current search depth, scope, and their interaction.
Controls and fixed effects	Current innovation scale and field-experience controls with firm and year effects.

DecisionPoint	LocalSpecification
Exclusion or substitution logic	The local executable design substitutes patent-based future outcomes for product-introduction outcomes and keeps only firm-years with usable lag-lead construction.

Expanded decision register for the manuscript.

D Appendix D: Sample Reconstruction

Statistic	Value
rows	2174.0000
unique_firms	162.0000
year_min	1986.0000
year_max	2015.0000
mean_search_depth	0.6637
mean_search_scope	4.2347
mean_next_patents	10.6026

Detailed sample reconstruction table.

E Appendix E: Variable Mapping

OriginalConstruct	LocalMeasure	ExactVariable	Derivation
Search depth	Mean prior organizational experience	search_depth	Average the current-year patent-level organizational experience within GVKEY-year.
Search scope	Mean breadth	search_scope	Average current-year breadth within GVKEY-year.
Current innovation scale	Current-year patent count	current_patents	Count current-year patents.
Next-period quantity	Next-year patent count	next_patents	Shift the patent count one year forward within GVKEY.
Next-period impact	Next-year total impact	next_total_impact	Shift the impact total one year forward within GVKEY.

Detailed construct mapping.

F Appendix F: Descriptive Statistics

Variable	N	Mean	SD	P25	Median	P75	Min	M
Search depth	2174.0000	0.6637	0.6622	0.1386	0.5493	0.9635	0.0000	4.76
Search scope	2174.0000	4.2347	2.8896	2.8182	3.4264	4.6667	2.0000	52.00
Current patents	2174.0000	10.5437	20.1122	2.0000	5.0000	11.0000	1.0000	399.00
Current total impact	2174.0000	56.5745	142.3809	5.0000	17.0000	47.0000	0.0000	3058.00
Mean field experience	2174.0000	1.4705	1.1482	0.6931	1.3041	1.9634	0.0000	7.55
Next-year patents	2174.0000	10.6026	21.2666	2.0000	4.0000	11.0000	1.0000	399.00
Next-year total impact	2174.0000	56.6297	144.9651	5.0000	16.0000	46.0000	0.0000	3058.00

Descriptive statistics appendix.

G Appendix G: Correlation Matrix

Variable	Search depth	Search scope	Current patents	Current total impact	Mean field experience	Next-year patents	Next-year total impact
Search depth	1.000	-0.011	0.184	0.138	0.735	0.158	0.106
Search scope	-0.011	1.000	0.010	0.146	-0.242	0.004	0.092
Current patents	0.184	0.010	1.000	0.612	0.062	0.746	0.513
Current total impact	0.138	0.146	0.612	1.000	0.042	0.482	0.670
Mean field experience	0.735	-0.242	0.062	0.042	1.000	0.046	0.020
Next-year patents	0.158	0.004	0.746	0.482	0.046	1.000	0.630
Next-year total impact	0.106	0.092	0.513	0.670	0.020	0.630	1.000

Correlation matrix for the main numeric variables.

H Appendix H: Yearly Audit

Period	V1	V2	V3	V4	V5
1986.0000	100.0000	0.2522	2.7740	7.7000	33.2900
1987.0000	103.0000	0.3220	2.9110	10.2136	45.3592
1988.0000	105.0000	0.3484	3.0106	13.1714	51.6571
1989.0000	98.0000	0.4390	3.1399	13.5102	61.0510
1990.0000	111.0000	0.4669	3.0152	11.9640	56.5225
1991.0000	109.0000	0.5001	3.1953	12.2110	61.1927
1992.0000	98.0000	0.4820	3.3594	13.5408	77.0510
1993.0000	100.0000	0.5504	3.4467	13.2200	82.5600
1994.0000	104.0000	0.5512	3.6508	11.6442	72.5481
1995.0000	98.0000	0.5892	3.7678	6.9490	45.8367
1996.0000	83.0000	0.6681	4.1041	7.6265	45.4217
1997.0000	80.0000	0.8318	4.0021	4.8500	30.2750
1998.0000	65.0000	0.6553	3.9114	9.2000	50.7077
1999.0000	78.0000	0.6504	3.7812	8.1410	45.8590
2000.0000	73.0000	0.8168	3.9949	8.3014	42.4384
2001.0000	72.0000	0.7542	4.2585	8.8056	42.2083
2002.0000	65.0000	0.8268	4.4429	8.0462	36.2308
2003.0000	62.0000	0.8369	4.8452	5.2581	19.5000
2004.0000	59.0000	0.7436	4.8832	5.0847	23.0678
2005.0000	48.0000	0.7782	5.1188	6.8125	33.2083
2006.0000	51.0000	0.8915	5.2182	4.9216	17.0392
2007.0000	43.0000	0.8791	6.2118	12.6047	48.6047
2008.0000	38.0000	0.9390	5.6773	5.2895	25.4737
2009.0000	43.0000	1.0399	6.1656	11.3488	49.2093
2010.0000	50.0000	0.9987	6.7648	12.9600	83.4800
2011.0000	51.0000	0.9412	7.1105	20.7059	83.2157

Period	V1	V2	V3	V4	V5
2012.0000	52.0000	1.0293	6.9358	16.5192	88.8462
2013.0000	50.0000	1.0361	6.7054	14.3400	180.3600
2014.0000	50.0000	1.1223	6.3393	21.3600	141.9400
2015.0000	35.0000	1.2033	6.8650	14.8857	57.1429

Year-by-year sample audit.

I Appendix I: Grouped Outcome Patterns

Quantile	N	MeanRegressor	MeanOutcome
1.0000	870.0000	0.1128	5.3149
2.0000	506.0000	0.5658	11.7273
3.0000	363.0000	0.8803	17.4105
4.0000	435.0000	1.6987	14.1885

Quantile summary for the main regressor.

J Appendix J: Full Coefficient Table

term	estimate	std_error	statistic	p_value	model
search_depth	-0.2243	0.1001	-2.2411	0.0250	katila_ahuja_next_pate
I(I(search_depth^2))	0.1059	0.0299	3.5451	0.0004	katila_ahuja_next_pate
search_scope	0.0052	0.0104	0.5025	0.6153	katila_ahuja_next_pate
log1p(current_patents)	0.6654	0.0652	10.2026	0.0000	katila_ahuja_next_pate
mean_field_exp	-0.0429	0.0262	-1.6360	0.1018	katila_ahuja_next_pate
search_depth:search_scope	-0.0115	0.0103	-1.1143	0.2651	katila_ahuja_next_pate
search_depth	-0.1070	0.1214	-0.8812	0.3854	katila_ahuja_next_impa
I(I(search_depth^2))	0.0388	0.0322	1.2061	0.2375	katila_ahuja_next_impa
search_scope	0.0184	0.0130	1.4170	0.1671	katila_ahuja_next_impa
log1p(current_patents)	0.6357	0.0617	10.3089	0.0000	katila_ahuja_next_impa
mean_field_exp	0.0271	0.0350	0.7753	0.4444	katila_ahuja_next_impa
search_depth:search_scope	-0.0216	0.0126	-1.7086	0.0982	katila_ahuja_next_impa

Full coefficient dump exported from the local estimation script.

K Appendix K: Missingness Audit

Variable	Missing	MissingShare
Search depth	0.0000	0.0000
Search scope	0.0000	0.0000
Current patents	0.0000	0.0000
Current total impact	0.0000	0.0000
Mean field experience	0.0000	0.0000
Next-year patents	0.0000	0.0000
Next-year total impact	0.0000	0.0000

Missingness audit for the variables summarized in the manuscript.

L Appendix L: Coverage of the Most Observed Entities

Entity	Observations	FirstPeriod	LastPeriod
10499.0000	30.0000	1986.0000	2015.0000
10983.0000	30.0000	1986.0000	2015.0000
1300.0000	30.0000	1986.0000	2015.0000
2285.0000	30.0000	1986.0000	2015.0000
2403.0000	30.0000	1986.0000	2015.0000
3532.0000	30.0000	1986.0000	2015.0000
4087.0000	30.0000	1986.0000	2015.0000
4321.0000	30.0000	1986.0000	2015.0000
4503.0000	30.0000	1986.0000	2015.0000
4839.0000	30.0000	1986.0000	2015.0000
5047.0000	30.0000	1986.0000	2015.0000
5439.0000	30.0000	1986.0000	2015.0000
6066.0000	30.0000	1986.0000	2015.0000
6266.0000	30.0000	1986.0000	2015.0000
7435.0000	30.0000	1986.0000	2015.0000
8762.0000	30.0000	1986.0000	2015.0000
2817.0000	29.0000	1986.0000	2015.0000
3835.0000	29.0000	1986.0000	2015.0000
6435.0000	29.0000	1986.0000	2015.0000
7257.0000	29.0000	1986.0000	2015.0000

Top entities by local observation count.

M Appendix M: Claim-Evidence Audit

Claim	EvidenceType	EvidenceFile	AuditNote
The source paper studies: The source paper asks whether search depth and search scope are distinct dimensions and how they shape subsequent innovation.	Direct source fact	patent literature/KatilaAhujaamj02.txt	Recovered from the source article before local writing begins.
The local executable panel contains 2,174 observations across 162 entities.	Local calculated fact	patent literature/daily_replications/2026-04-04/katila-ahuja-2002-search-behavior/results/katila_ahuja_2002_analysis_dataset.cs — patent literature/daily_replications/2026-04-04/katila-ahuja-2002-search-behavior/results/katila_ahuja_2002_sample_summary.c	Backed by the verified analysis panel and sample audit.
The local headline estimate is 0.0052 for search_scope with p = 0.6153.	Local calculated fact	patent literature/daily_replications/2026-04-04/katila-ahuja-2002-search-behavior/results/katila_ahuja_2002_model_coefficients	Backed by the exported coefficient file used in the paper.

Claim	EvidenceType	EvidenceFile	AuditNote
The manuscript uses local substitutions or analogues wherever the original data architecture is not fully recoverable.	Replication decision	Source-vs-decision table and limitations section	Used only when the original pipeline cannot be matched byte-for-byte from current local files.

Expanded claim-evidence audit.

N Appendix N: Source Inventory Note

Source Inventory - paper: Katila and Ahuja (2002) - source title: Something Old, Something New: A Longitudinal Study of Search Behavior and New Product Introduction - source file: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/KatilaAhujaamj02.txt‘ ## Paper-First Reading Protocol - Step 1: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/KatilaAhujaamj02.txt‘ — needed for: Read the original paper before touching the local estimates. — exists: n/a — status: workflow step - Step 2: ‘Something Old, Something New: A Longitudinal Study of Search Behavior and New Product Introduction‘ — needed for: Extract the source paper’s theoretical logic, sample definition, variable definitions, and estimand. — exists: n/a — status: workflow step - Step 3: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/katila-ahuja-2002-search-behavior/results/katila_ahuja_2002_analysis_dataset.csv — C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/katila-ahuja-2002-search-behavior/results/katila_ahuja_2002_model_coefficients.csv — C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/katila-ahuja-2002-search-behavior/results/katila_ahuja_2002_sample_summary.csv‘ — needed for: Match each required empirical object to a local executable file and verify that the file exists. — exists: n/a — status: workflow step - Step 4: ‘The manuscript, markdown notes, and PDF are generated from the verified file set.‘ — needed for: Write the replication paper only from the checked local files and calculated outputs. — exists: n/a — status: workflow step ## Local files - Source article: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/KatilaAhujaamj02.txt‘ — needed for: Recover theory, hypotheses, sample rules, definitions, tables, notes, and appendices — exists: Yes — status: Direct source - Analysis panel: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/katila-ahuja-2002-search-behavior/results/katila_ahuja_2002_analysis_dataset.c — needed for: Estimate the local model and regenerate descriptive statistics — exists: Yes — status: Executable local dataset - Coefficient export: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/katila-ahuja-2002-search-behavior/results/katila_ahuja_2002_model_coefficients.csv‘ — needed for: Recover the re-

ported local coefficients and p-values — exists: Yes — status: Executable local result file - Sample summary: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/katila-ahuja-2002-search-behavior/results/katila_ahuja_2002_sample_summary.c — needed for: Audit sample construction and raw grouped moments — exists: Yes — status: Executable local result file

O Appendix O: Study Map Note

Study Map - research question: The source paper asks whether search depth and search scope are distinct dimensions and how they shape subsequent innovation. - positioning: The replication matters because the original paper's distinction between search depth and search scope is widely cited, yet the empirical mechanism depends heavily on how those constructs are operationalized. - contribution: The paper's contribution is to recover the two-dimensional search logic in an executable patent panel and show where the patent-based analogue departs from the original product-outcome result. - local design: The local analogue aggregates the patent-level recombination panel to firm-years and uses patent-based outcomes in place of the source paper's product-introduction outcomes. - estimand: Firm and year fixed-effects models for next-year patent counts and next-year total impact using local depth, scope, and their interaction. - headline estimate: '0.0052' with 'p = 0.6153'. ## Decision Audit - Research question: The source paper asks whether search depth and search scope are distinct dimensions and how they shape subsequent innovation. - Unit of analysis: Firm-year - Sample frame: Firm-year panel aggregated from the local patent-level recombination dataset for firms with usable year-to-year transitions. - Time window: 1986 to 2015 in the local executable panel. - Outcome definition: Next-year patent quantity and next-year total patent impact. - Main explanatory design: Firm and year fixed-effects models relating next-period innovation to current search depth, scope, and their interaction. - Controls and fixed effects: Current innovation scale and field-experience controls with firm and year effects. - Exclusion or substitution logic: The local executable design substitutes patent-based future outcomes for product-introduction outcomes and keeps only firm-years with usable lag-lead construction. ## Source Versus Decision - direct source fact: Depth and scope are distinct dimensions. - direct source fact: The original design is firm-year based. - direct source fact: Innovation outcomes include quantity and quality. local replication decision: The local panel maps depth and scope to executable recombination constructs. local replication decision: The firm-year structure is preserved exactly. local replication decision: Patent counts and patent-impact totals substitute for product introductions.

P Appendix P: Data Design Note

Data Design - Firm-year panel aggregated from the local patent-level recombination dataset. - Search depth measured from prior organizational recombination experience. - Search scope measured from breadth of recombination in the current year. - panel file used: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/katila-ahuja-2002-search-behavior/results/katila_ahuja_2002_analysis_dataset.csv‘ - coefficient file used: ‘C:/Users/bai/On - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/katila-ahuja-2002-search-behavior/results/katila_ahuja_2002_model_coefficients.csv‘ - sample summary used: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026 04-04/katila-ahuja-2002-search-behavior/results/katila_ahuja_2002_sample_summary.csv‘ - observations: ‘2,174‘ - entities: ‘162‘ ## Required Data And Local File Mapping - Source article: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/KatilaAhujaamj02.txt‘ — needed for: Recover theory, hypotheses, sample rules, definitions, tables, notes, and appendices — exists: Yes — status: Direct source - Analysis panel: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/katila-ahuja-2002-search-behavior/results/katila_ahuja_2002_analysis_dataset.csv‘ — needed for: Estimate the local model and regenerate descriptive statistics — exists: Yes — status: Executable local dataset - Coefficient export: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/katila-ahuja-2002-search-behavior/results/katila_ahuja_2002_model_coeffic — needed for: Recover the reported local coefficients and p-values — exists: Yes — status: Executable local result file - Sample summary: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/katila-ahuja-2002-search-behavior/results/katila_ahuja_2 — needed for: Audit sample construction and raw grouped moments — exists: Yes — status: Executable local result file

Q Appendix Q: Derivations Note

Derivations And Estimands - The local quantity model points to a nonlinear depth effect and a weaker role for search scope than the source paper emphasizes. - Firm and year fixed-effects models for next-year patent counts and next-year total impact using local depth, scope, and their interaction. - model equations are reproduced in ‘paper/main.tex‘. ## Variable Map - Search depth -¿ ‘search_depth‘ : Average the current-year patent-level organizational experience within GVKEY-year. - Search scope -¿ ‘search_scope‘ : Average current-year breadth within GVKEY-year. - Current innovation scale -¿ ‘current_patents‘ : Count current-year patents. - Next-period quantity -¿ ‘next_patents‘ : Shift the patent count one year forward within GVKEY. - Next-period impact -¿ ‘next_total_impact‘ : Shift the impact total one year forward within GVKEY. ## Claim Evidence Audit - Direct source fact: The source paper studies: The source paper asks whether search depth and search scope are distinct dimensions and how they shape subsequent innovation. — evidence: patent literature/KatilaAhujaamj02.txt - Local calculated fact: The local executable panel contains 2,174 observations across 162 entities. — evidence: patent literature/daily_replications/2026-04-04/katila-ahuja-2002-search-behavior/results/katila_ahuja_2002_analysis_dataset.csv — patent literature/daily_replications/2026-04-04/katila-ahuja-2002-search-behavior/results/katila_ahuja_2002_sample_summa - Local calculated fact: The local headline estimate is 0.0052 for search_scope with p = 0.6153. — evidence: patent literature/daily_replications/2026-04-04/katila-ahuja-2002-search-behavior/results/katila_ahuja_200 - Replication decision: The manuscript uses local substitutions or analogues wherever the original data architecture is not fully recoverable. — evidence: Source-vs-decision table and limitations section

R Appendix R: Replication Decisions Note

Replication Decisions - The local analogue uses patent outcomes rather than product-introduction outcomes. - Search variables are mapped from recombination measures already in the workspace. - The exercise is strongest on conceptual mechanism and panel structure, not exact outcome identity. - direct source facts are separated from local replication decisions in the manuscript. - no coefficient or table is fabricated. ## Source Facts Versus Local Decisions - source: Depth and scope are distinct dimensions. - source: The original design is firm-year based. - source: Innovation outcomes include quantity and quality. decision: The local panel maps depth and scope to executable recombination constructs. decision: The firm-year structure is preserved exactly. decision: Patent counts and patent-impact totals substitute for product introductions.

S Appendix S: Local Execution Status Note

Local Execution Status - batch date: ‘2026-04-04‘ - status: long-form journal-style manuscript generated from local result files - current paper: ‘katila-ahuja-2002-search-behavior‘

T Appendix T: Cleaned Output Tables Note

Cleaned Output Tables - paper-first reading protocol - required-data and file-manifest table - decision register - sample reconstruction table - local source inventory table - source facts versus replication decisions table - claim-evidence audit table - variable mapping table - key coefficient table - full coefficient table - descriptive statistics - correlation matrix - yearly audit - missingness audit - top-entity coverage table - quantile summary

U Appendix U: Build Notes

Build Notes

- batch date: ‘2026-04-04‘ - manuscript style: long-form journal-style replication draft - evidence rule: only processed local datasets and actual local estimates are used - workflow rule: read the source paper first, then map required datasets to verified local files, then write the paper from those files - skill path guiding the structure: ‘C:/Users/bai/.codex/skills/write-replication-journal-github/SKILL.md‘

V Appendix V: Manuscript Metadata

Field	Value
Paper	Katila and Ahuja (2002)
Source title	Something Old, Something New: A Longitudinal Study of Search Behavior and New Product Introduction
Authors	Riitta Katila and Gautam Ahuja
Outlet	Academy of Management Journal
Batch date	2026-04-04
Rows	2,174
Entities	162
Headline term	search_scope
Headline estimate	0.0052
Headline p-value	0.6153
Panel path	C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/katila-ahuja-2002-search-behavior/results/katila_ahuja_2002_analysis_dataset.csv

Metadata block for GitHub publishing and future reruns.

W Appendix W: Replication Summary Note

Research question: The source paper asks whether search depth and search scope are distinct dimensions and how they shape subsequent innovation. Local design: The local analogue aggregates the patent-level recombination panel to firm-years and uses patent-based outcomes in place of the source paper's product-introduction outcomes. Estimand: Firm and year fixed-effects models for next-year patent counts and next-year total impact using local depth, scope, and their interaction. Local result focus: The local quantity model points to a nonlinear depth effect and a weaker role for search scope than the source paper emphasizes. Comparison note: The local analogue preserves the paper's dual-dimension logic but does not produce the same relative strength for depth and scope. The quantity model highlights a nonlinear depth pattern, while search scope is comparatively muted in the executable patent-based panel.

References

[1] Riitta Katila and Gautam Ahuja. Something old, something new: A longitudinal study of search behavior and new product introduction. *Academy of Management Journal*, 45(6):1183–1194, 2002.